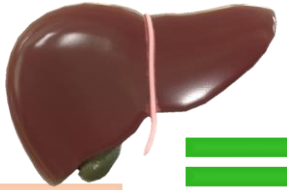
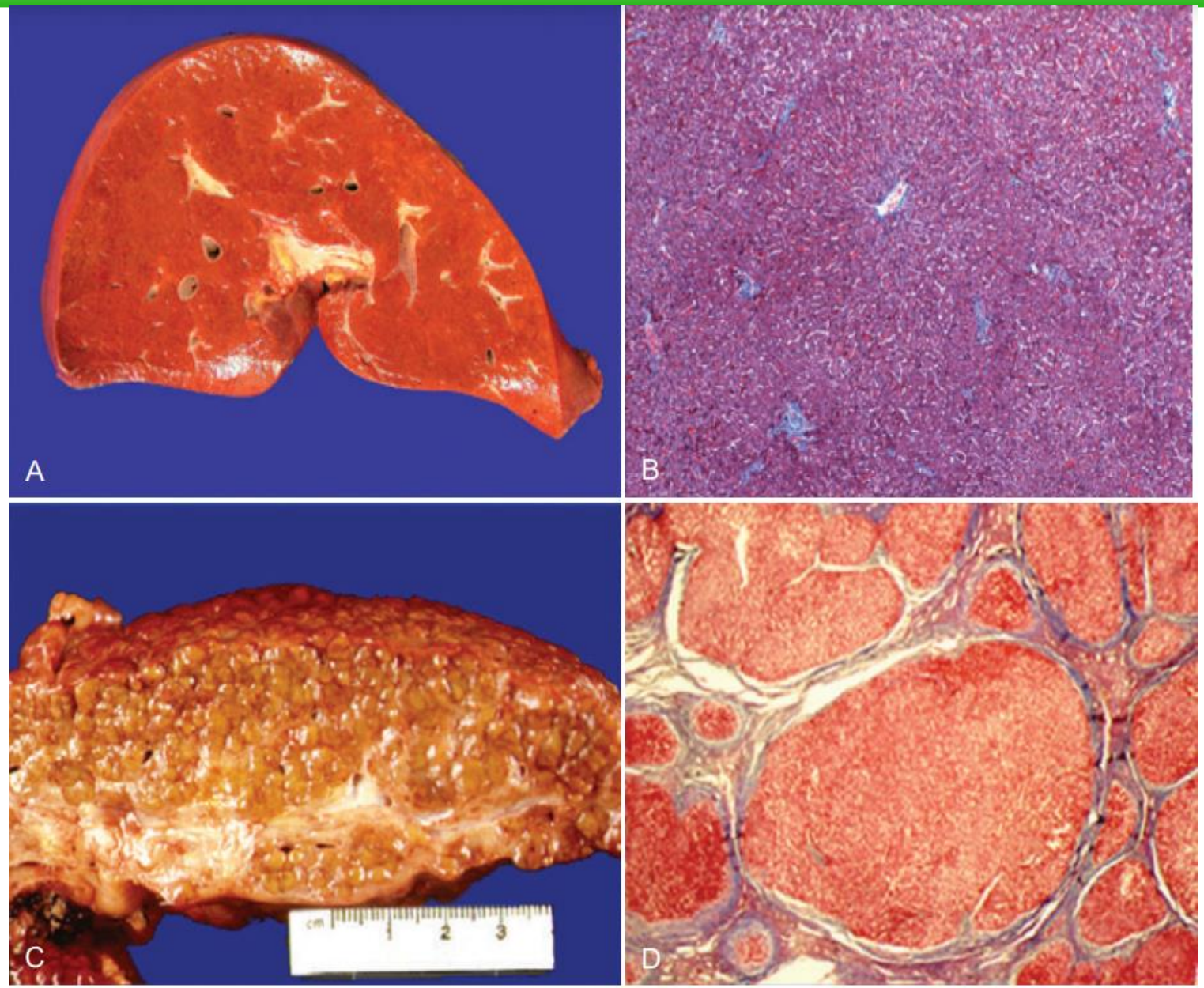
CIRRHOSIS AND PORTAL HYPERTENSION



Introduction

- Cirrhosis is a diffuse process characterized by fibrosis and conversion of normal architecture to structurally abnormal nodules ("regenerative" nodules) which lack normal lobular organization
- Cirrhosis → Final stage of any chronic liver disease
- Histopathological diagnosis → all-or-nothing
- Clinically
    1. Compensated cirrhosis
    2. Decompensated cirrhosis → Ascites, variceal hemorrhage, encephalopathy or jaundice

Histopathology



Causes

1. Chronic hepatitis C
2. Alcoholic liver disease
3. Non alcoholic fatty liver disease(NAFLD)
4. Chronic hepatitis B

"Cryptogenic cirrhosis"

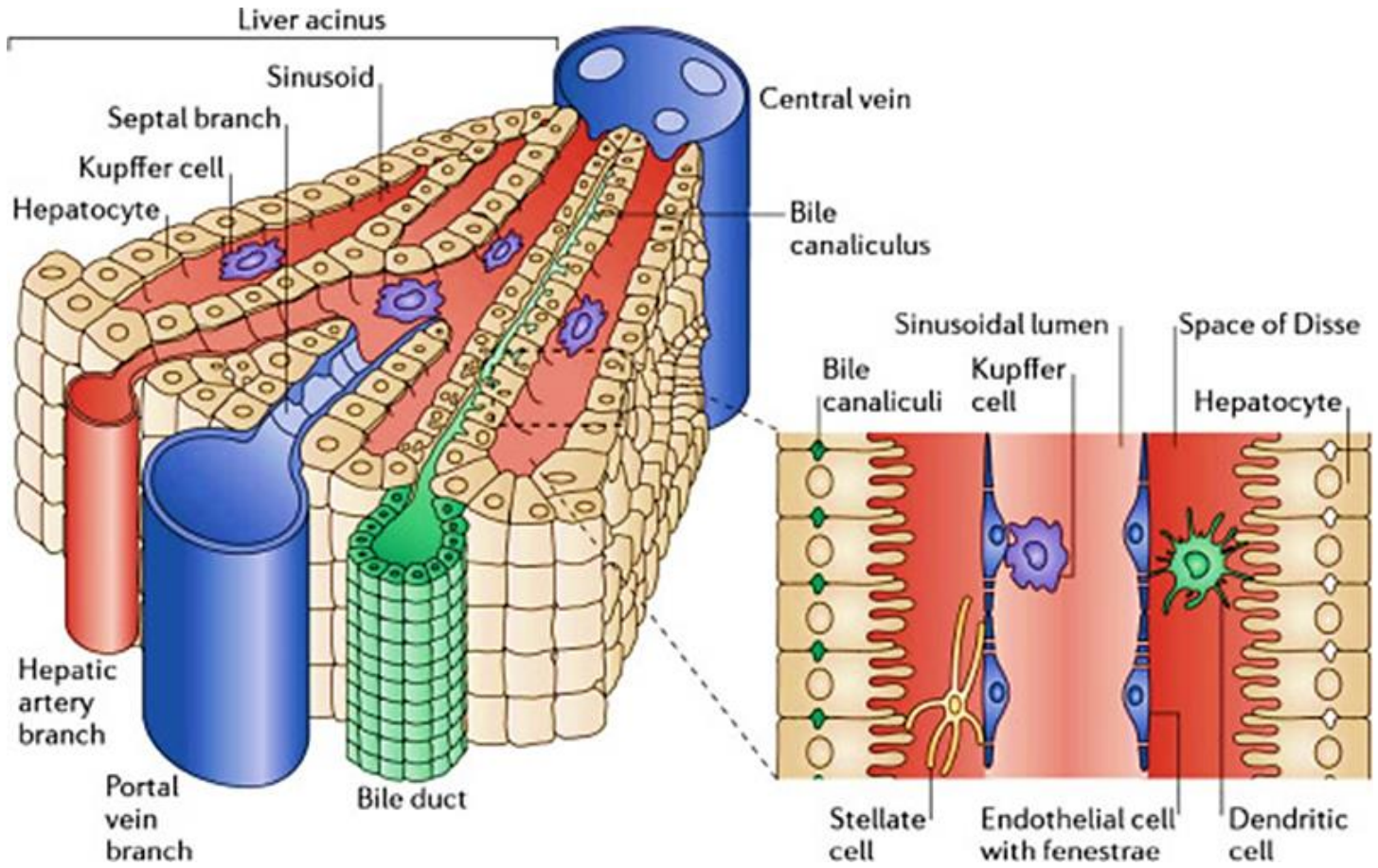
In India → MC cause – Alcohol f/b Hepatitis B

TABLE 153-1 CAUSES OF CIRRHOSIS

MAIN FACTORS CAUSING CIRRHOSIS
Chronic hepatitis C
Alcoholic liver disease
Nonalcoholic fatty liver disease
Chronic hepatitis B
OTHER CAUSES OF CIRRHOSIS (<2% OF ALL CASES)
Cholestatic and autoimmune liver diseases
Primary biliary cirrhosis
Primary sclerosing cholangitis
Autoimmune hepatitis
Intrahepatic or extrahepatic biliary obstruction
Mechanical obstruction
Biliary atresia
Cystic fibrosis
Metabolic disorders
Hemochromatosis
Wilson disease
α_1-Antitrypsin deficiency
Glycogen storage diseases
Abetalipoproteinemia
Porphyria
Hepatic venous outflow obstruction
Budd-Chiari syndrome
Veno-occlusive disease
Right-sided heart failure
Drugs and toxins
Intestinal bypass
Indian childhood cirrhosis

Pathobiology

- Key pathogenic feature → activation of hepatic stellate cells (Ito cells or perisinusoidal cells) located in the space of Disse between hepatocytes and sinusoidal endothelial cells
- Injury → hepatic stellate cells become activated → lose their vitamin A deposits →proliferate, develop a prominent rough endoplasmic reticulum → secrete extracellular matrix (collagen types I and III, sulfated proteoglycans, and glycoproteins) → become contractile hepatic myofibroblasts


Liver acinus
Sinusoid
Septal branch
Kupffer cell
Hepatocyte
Central vein
Bile canaliculus
Hepatic artery branch
Portal vein branch
Bile duct
Sinusoidal lumen
Space of Disse
Bile canaliculi
Kupffer cell
Hepatocyte
Stellate cell
Endothelial cell with fenestrae
Dendritic cell

Pathobiology

- Unlike other capillaries, normal hepatic sinusoids lack a basement membrane
- sinusoidal endothelial cells themselves contain large fenestrae
- Collagen deposition in the space of Disse → defenestration of the sinusoidal endothelial cells ("capillarization" of the sinusoids) → altering exchange between plasma and hepatocytes → resulting in a decreased sinusoidal diameter that is further exacerbated by the contraction of stellate cells

Complications of Cirrhosis

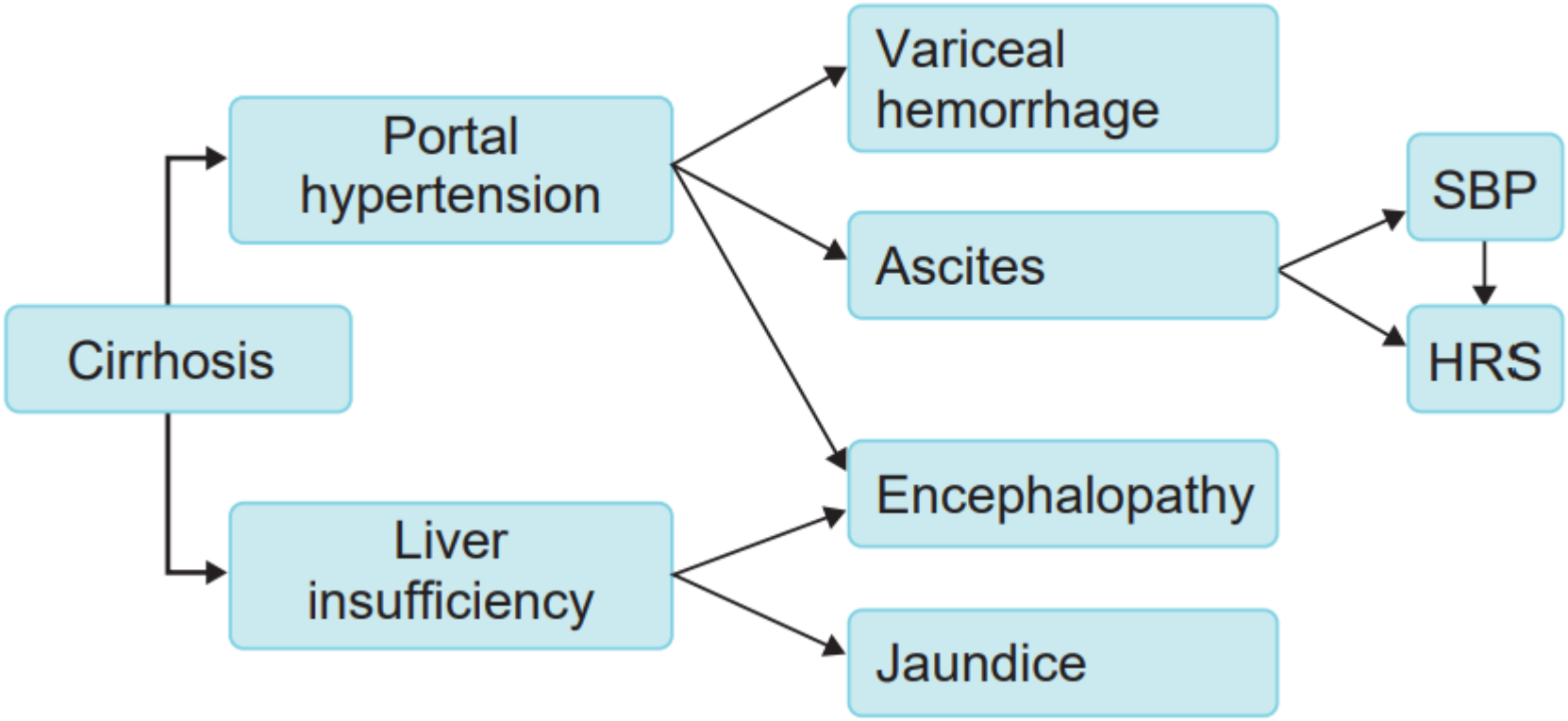
1. Portal hypertension
2. Liver insufficiency

A. Varices and Ascites → due to portal hypertension

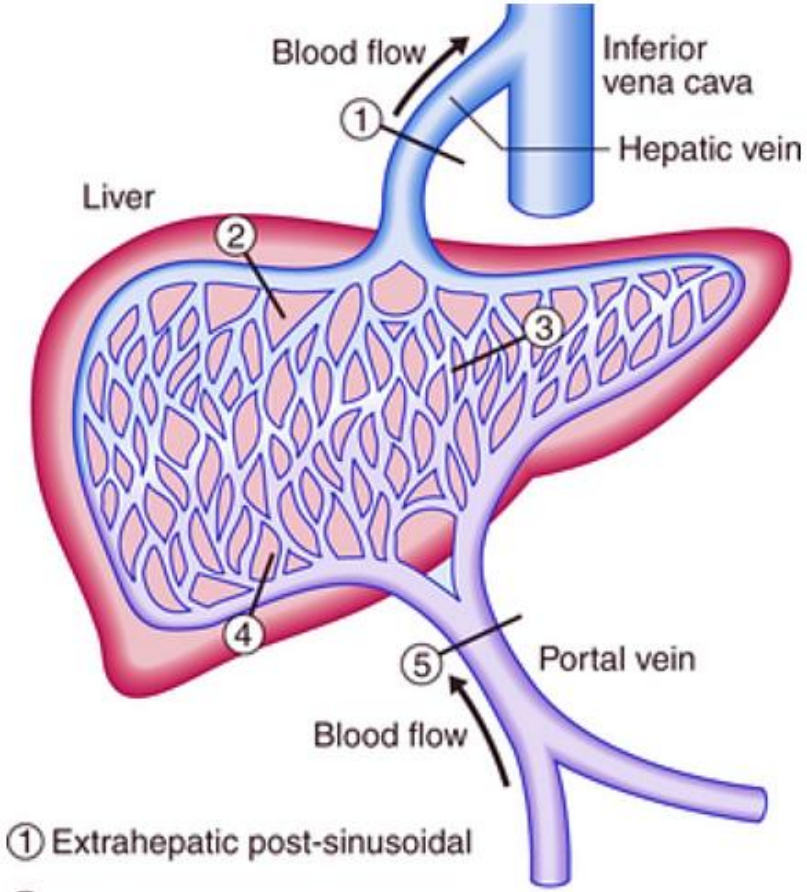
B. Jaundice → due to liver insufficiency

C. Encephalopathy → due to both portal hypertension and liver insufficiency

Complications of Cirrhosis



Portal Hypertension - Classification



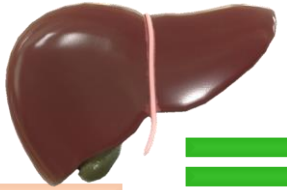
① Extrahepatic post-sinusoidal

② Intrahepatic post-sinusoidal

③ Sinusoidal

④ Intrahepatic pre-sinusoidal

⑤ Extrahepatic pre-sinusoidal



Portal Hypertension - Causes

PREHEPATIC

- Portal vein thrombosis
- Splenic vein thrombosis
- Massive splenomegaly (Banti's syndrome)

HEPATIC

- **Presinusoidal**
  - Schistosomiasis
  - Congenital hepatic fibrosis
- **Sinusoidal**
  - Cirrhosis—many causes
  - Alcoholic hepatitis
- **Postsinusoidal**
  - Hepatic sinusoidal obstruction (venoocclusive syndrome)

POSTHEPATIC

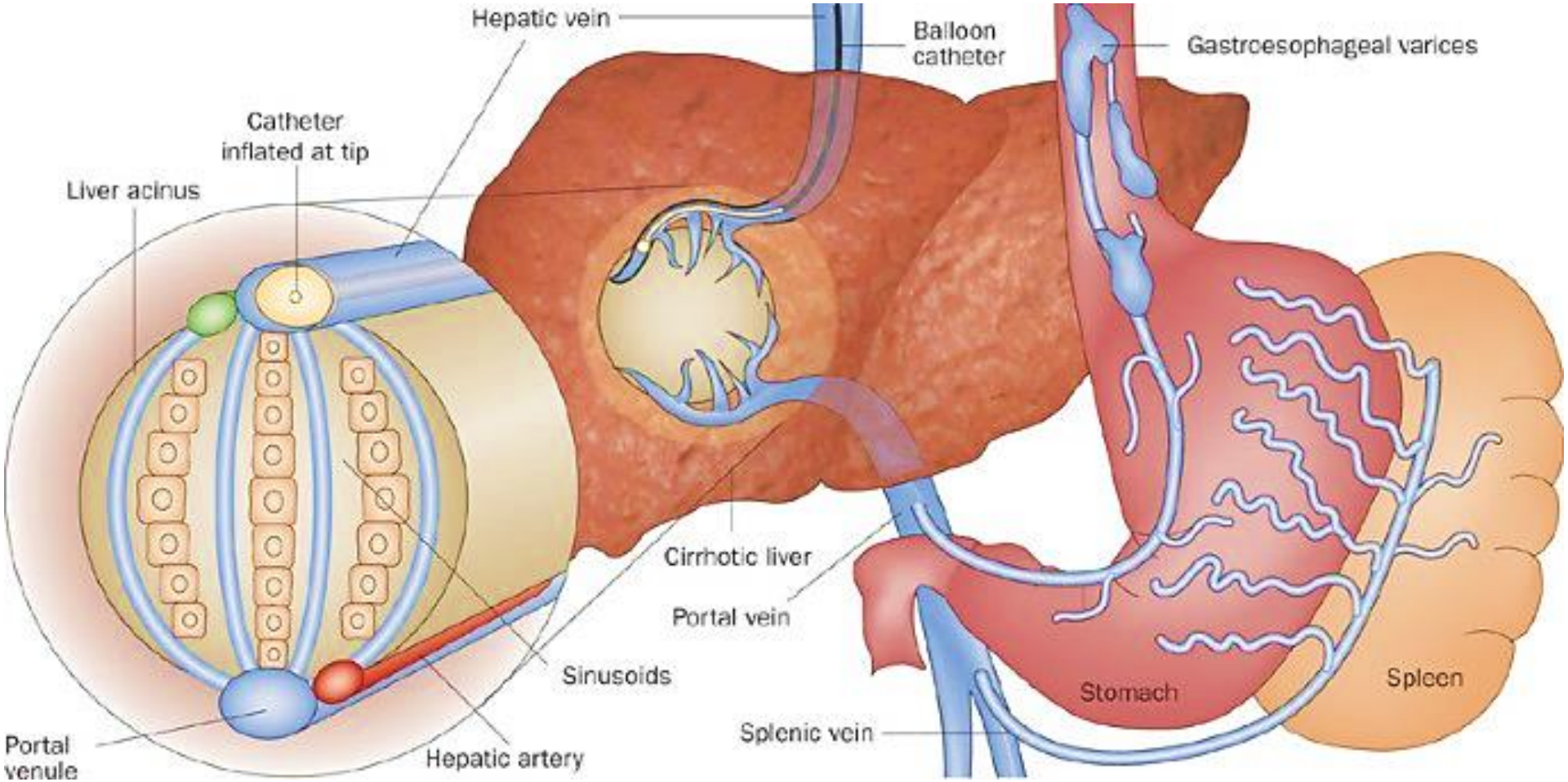
- Budd-Chiari syndrome
- Inferior vena caval webs
- Cardiac causes
  - Restrictive cardiomyopathy
  - Constrictive pericarditis
  - Severe congestive heart failure

Portal hypertension is defined as the elevation of the hepatic venous pressure gradient (HVPG) to >5 mmHg

Portal pressure Measurements

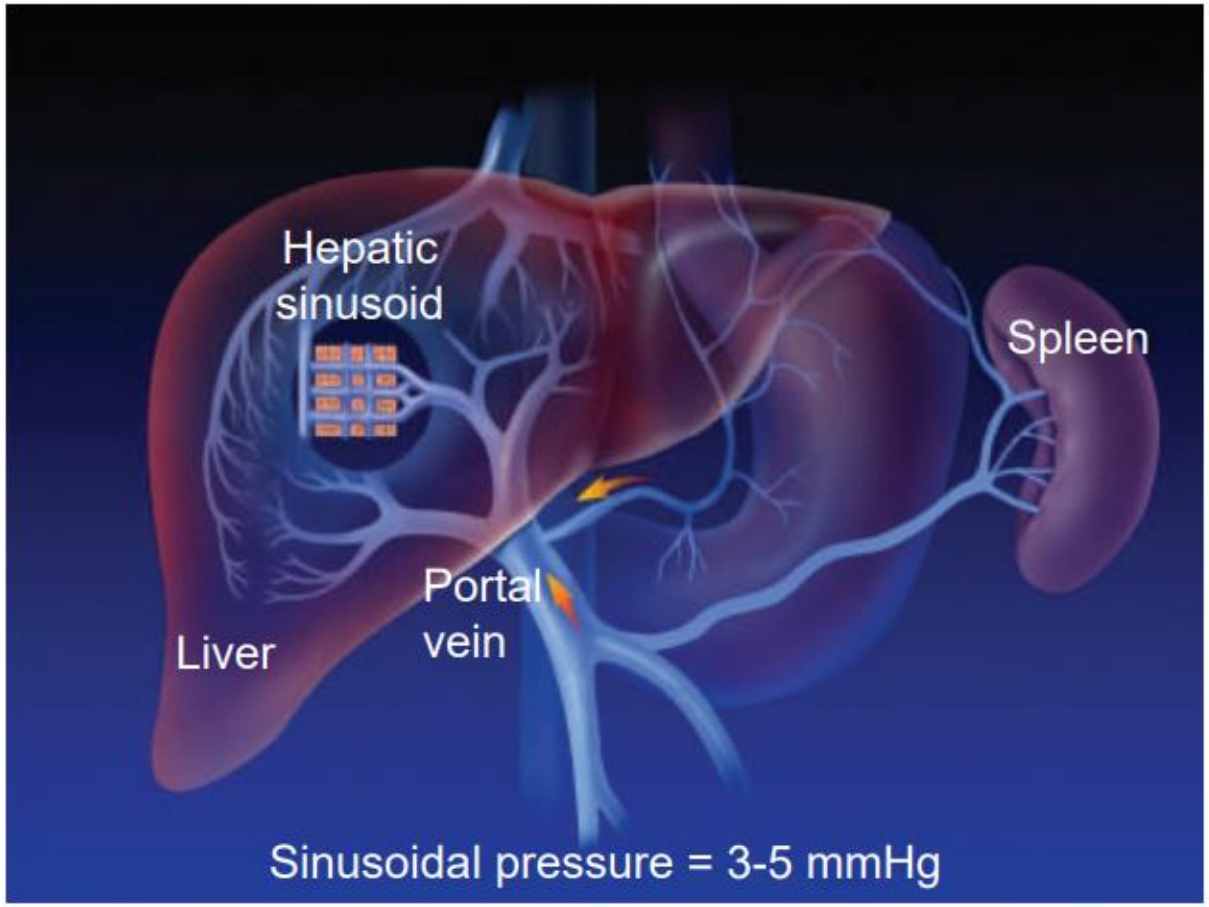
- Portal pressure measurements are expressed as the hepatic venous pressure gradient: the gradient between wedged hepatic venous pressure, which is a measure of sinusoidal pressure, and free hepatic or inferior vena cava pressure, which is used as an internal zero reference point
- HVPG normal (3 to 5 mmHg) → Prehepatic and intrahepatic presinusoidal
- HVPG elevated (> 5 mmHg) → Sinusoidal and posthepatic portal hypertension
- Hepatic venous pressure gradient of 10 mm Hg or greater → “clinically significant” portal hypertension

HVPG

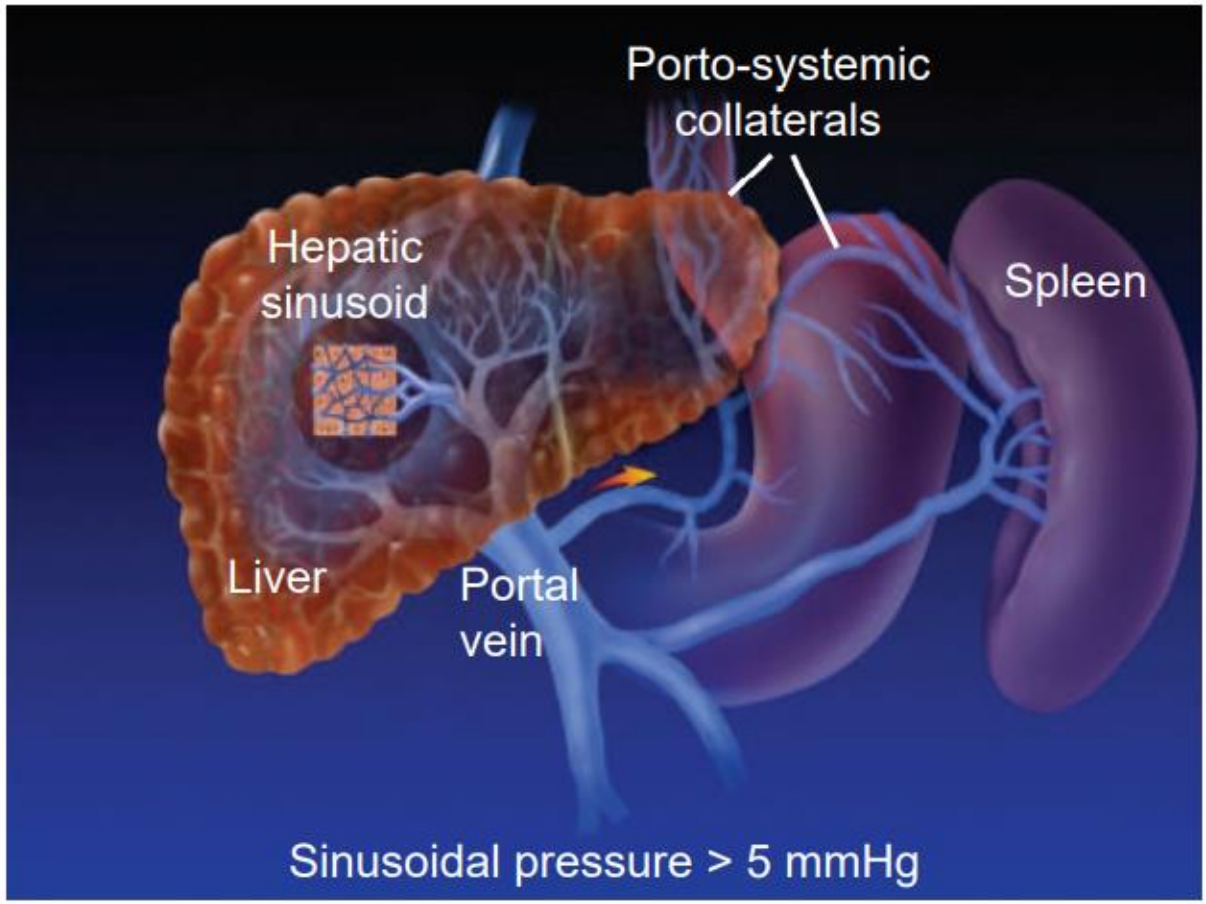

Hepatic vein
Balloon catheter
Gastroesophageal varices
Catheter inflated at tip
Liver acinus
Cirrhotic liver
Portal vein
Sinusoids
Stomach
Spleen
Splenic vein
Portal venule
Hepatic artery

Portal Hypertension

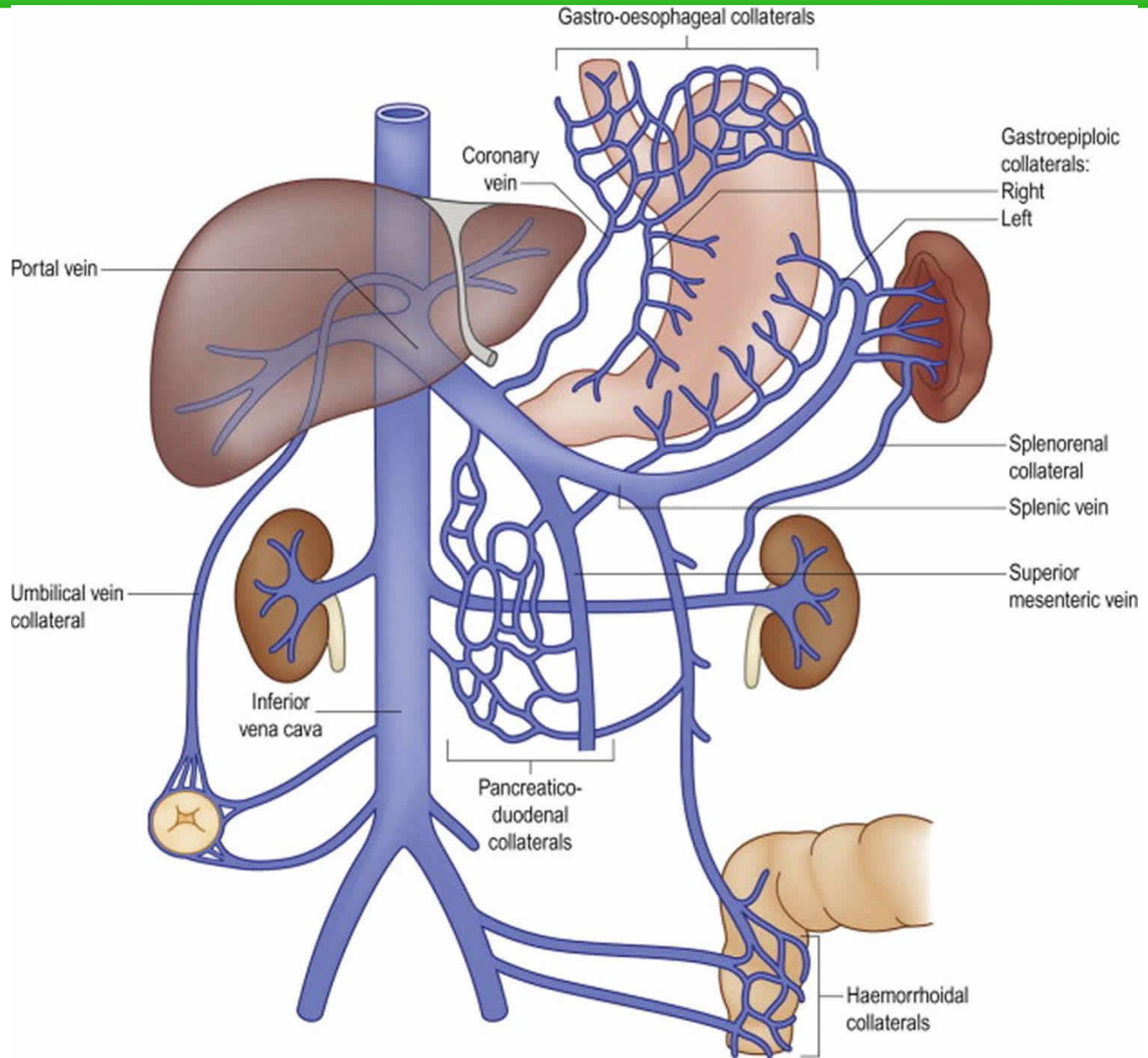
Normal



Cirrhosis



Collaterals



Portal Hypertension In Cirrhosis

1. **Increased in resistance to portal flow**
   A. Deposition of fibrous tissue and subsequent compression by regenerative nodules (fixed component)
   B. Active vasoconstriction (functional component) → due to deficiency of intrahepatic nitric oxide
2. **Increase in portal venous inflow**
   1. Splanchnic vasodilation (due to increased NO)
   2. Collaterals

Paradox in portal hypertension

→ *Deficiency of NO in the intrahepatic vasculature leads to vasoconstriction and increased resistance*

→ *overproduction of NO in the extrahepatic circulation leads to vasodilation and increased portal flow*

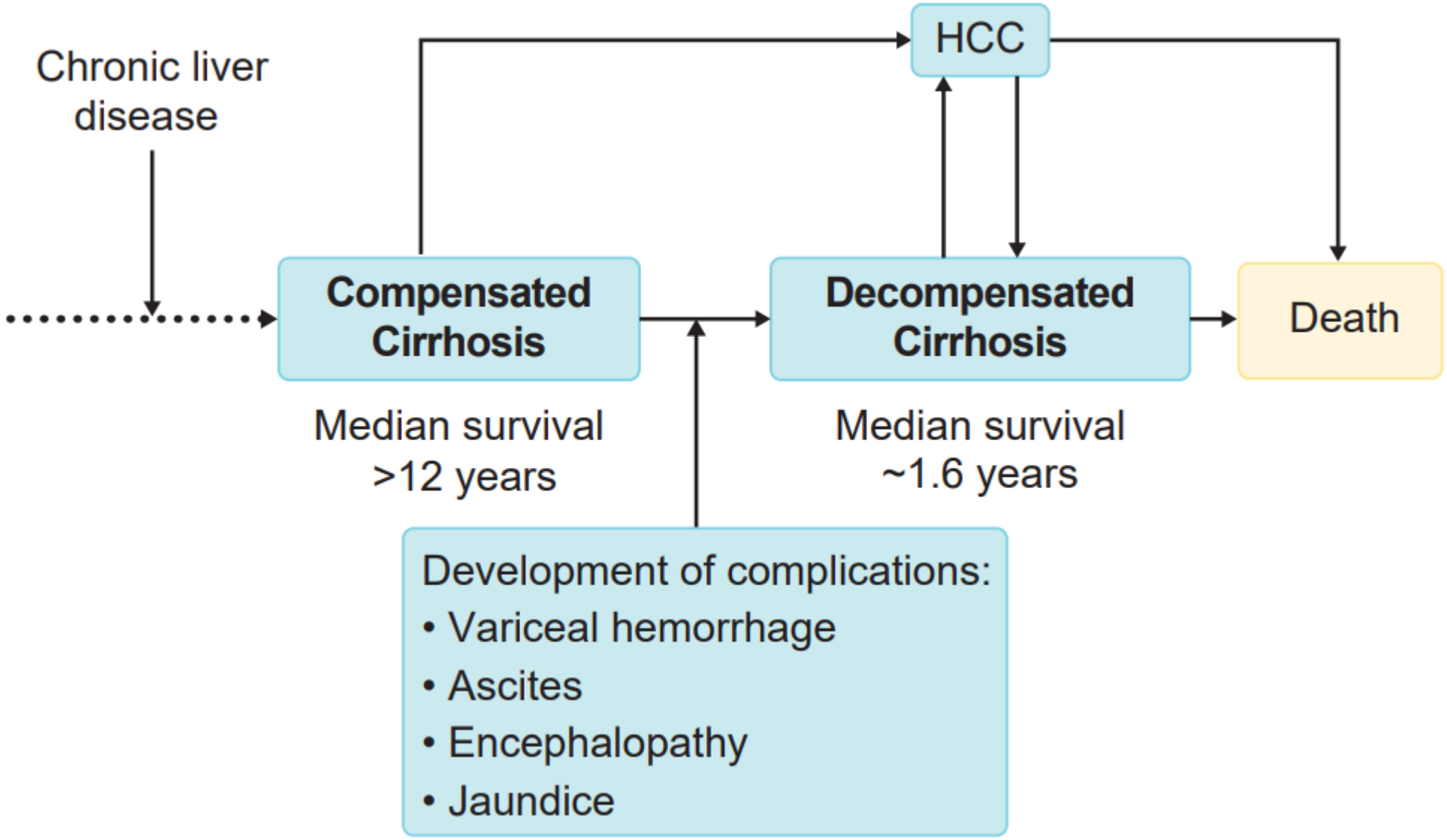
Hyperdynamic circulatory state

- Systemic vasodilation → decreased effective arterial blood volume → activation of neurohumoral systems (renin-angiotensin-aldosterone system) → retention of sodium, expansion of plasma volume, and development of a hyperdynamic circulatory state
- Hyperdynamic circulatory state maintains portal hypertension → leading to the formation and growth of varices →development of all other complications of cirrhosis

Clinical Manifestations

- Compensated cirrhosis → liver synthetic function is normal, portal pressure is increased but below the threshold for development of varices or ascites
- Decompensated cirrhosis → ascites, portal hypertensive gastrointestinal bleeding, encephalopathy, jaundice
- Transition from a compensated to a decompensated stage occurs at a rate of approximately 5 to 7% per year.
- The median time to decompensation, or the time at which half the patients with compensated cirrhosis will become decompensated, is about 6 years.

Natural History Of Cirrhosis



Compensated Cirrhosis

- Mostly asymptomatic
- Diagnosed during biochemical testing, imaging, endoscopy showing varices
- 40% patients with compensated cirrhosis have esophageal varices (without bleeding)

Decompensated Cirrhosis

1. Ascites
2. Variceal haemorrhage
3. Jaundice
4. Hepatic encephalopathy

- Ascites → most frequent sign of decompensation → present in 80% of patients with decompensated cirrhosis

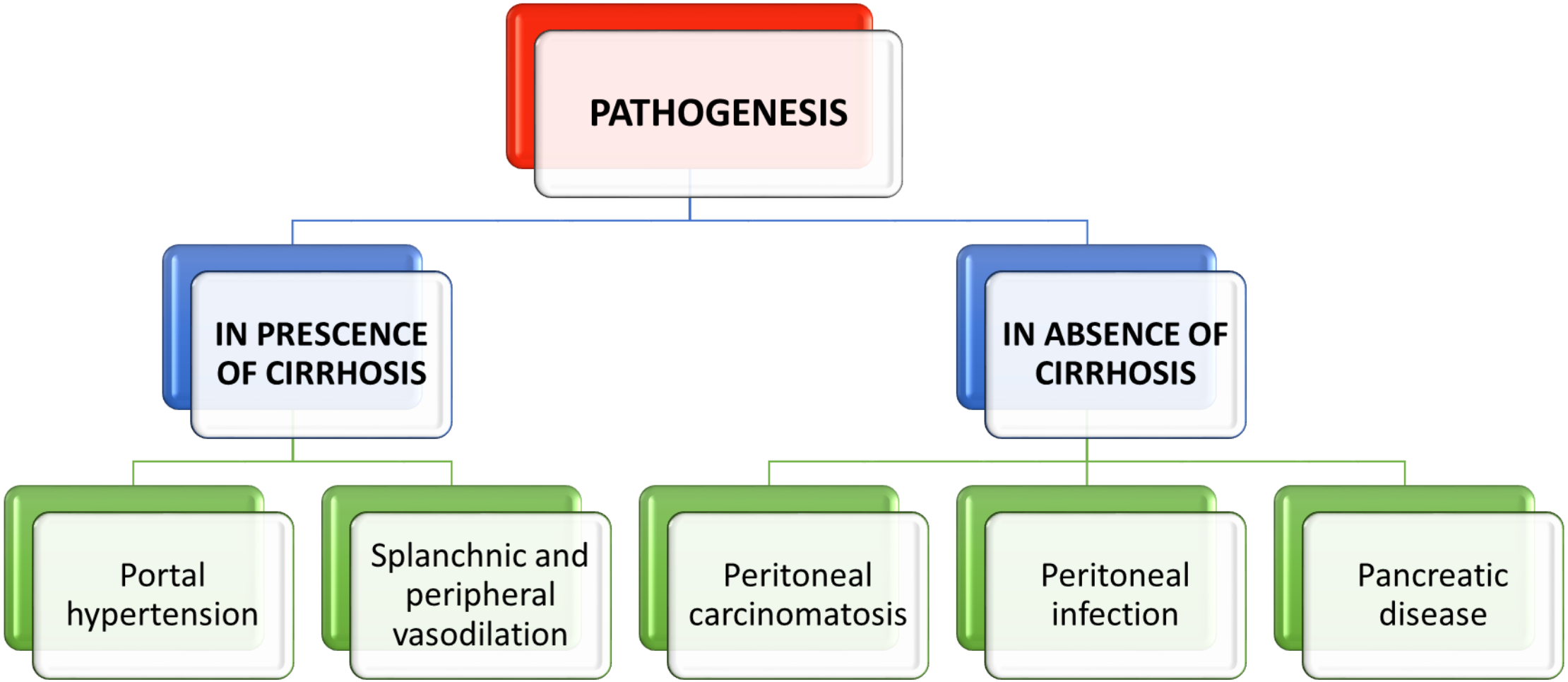
Variceal hemorrhage

- Gastroesophageal varices are present in approximately 50% of patients with newly diagnosed cirrhosis
- Prevalence of varices → 40% in Child A cirrhotic patients to 85% in Child C cirrhotic patients
- Both the development of varices and the growth of small varices occur at a rate of 7 to 8% per year
- Portal-systemic collaterals, the most relevant of which are those that form through dilation of the coronary and gastric veins and constitute gastroesophageal varices
- Hepatic venous pressure gradient of 10 to 12 mm Hg, below which varices do not develop.

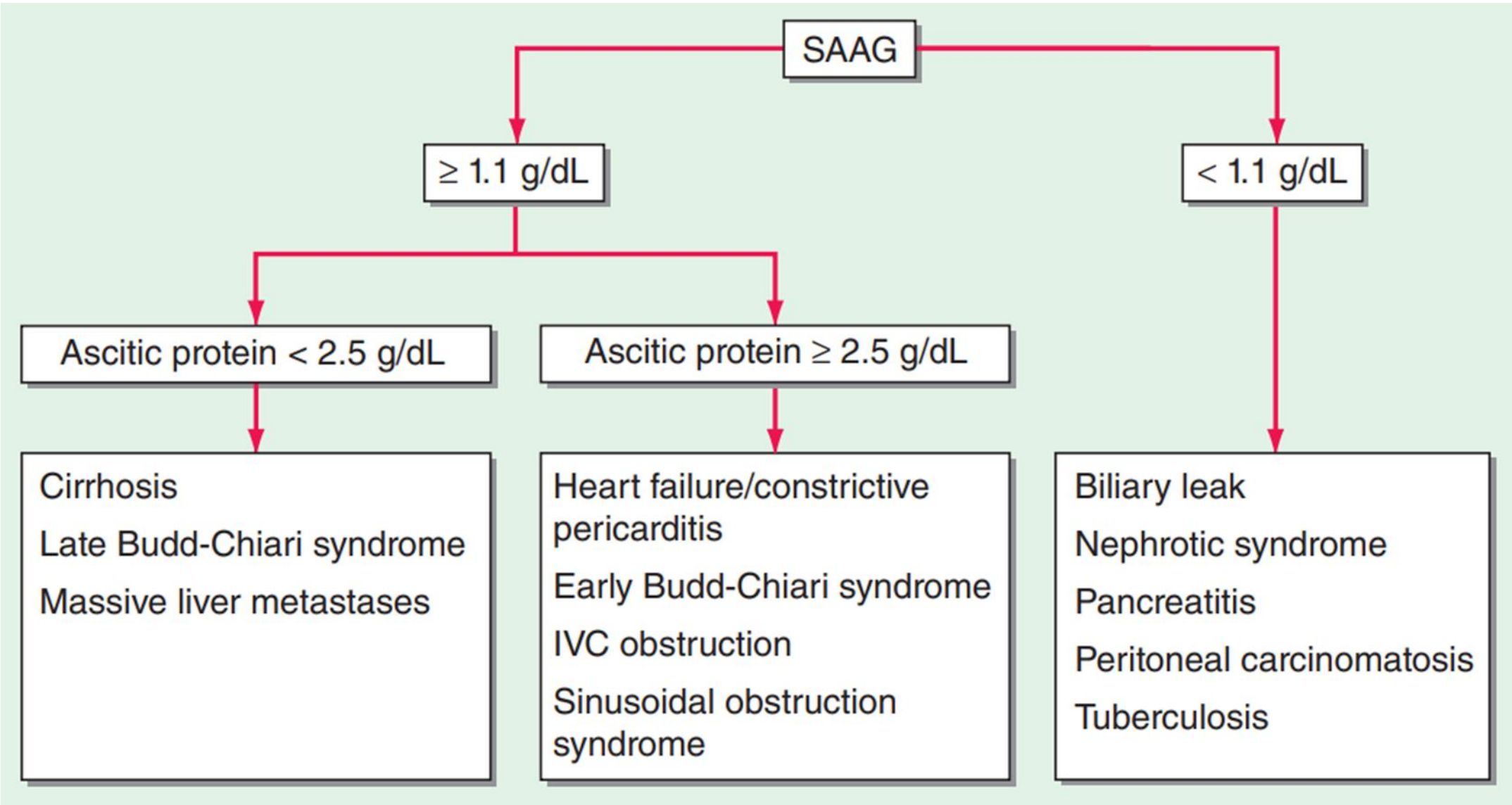
Ascites

- 7 to 10% per year
- Secondary to sinusoidal hypertension and retention of sodium
- Threshold HVPG of 12 mm Hg is needed for the formation of ascites

Ascites


PATHOGENESIS
IN PRESCENCE OF CIRRHOSIS
IN ABSENCE OF CIRRHOSIS
Portal hypertension
Splanchnic and peripheral vasodilation
Peritoneal carcinomatosis
Peritoneal infection
Pancreatic disease

Ascites - SAAG



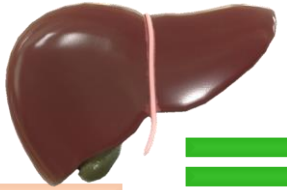
SAAG and Ascitic Protein

- Protein levels in ascitic fluid are an indirect marker of the integrity of the hepatic sinusoids: normal sinusoids are permeable structures that “leak” protein, whereas sinusoids in cirrhosis are “capillarized” and do not leak as much protein

TABLE 153-3 USING THE SERUM-ASCITES ALBUMIN GRADIENT AND THE ASCITES TOTAL PROTEIN LEVEL TO DIAGNOSE THE CAUSE OF ASCITES

CONDITION	SERUM-ASCITES ALBUMIN GRADIENT*	ASCITES TOTAL PROTEIN LEVEL†
Cirrhosis	High	Low
Malignant ascites	Low	High
Cardiac ascites	High	High

*High is more than 1.1 g/dL; low is less than 1.1 g/dL.
†High is more than 2.5 g/dL; low is less than 2.5 g/dL.



Causes Of ascites

	High SAAG (Transudative)	Low SAAG (Exudative)
Common causes	1. **Hepatic cirrhosis** 2. Cardiac failure	1. **Malignant disease** a. Hepatic b. Peritoneal
Other causes	3. Hyperproteinemia → Protein losing enteropathy → Malnutrition 4. Hepatic venous occlusion → Budd-Chiari syndrome → Sinusoidal obstruction syndrome	2. Acute pancreatitis 3. Lymphatic obstruction 4. Infection → Tuberculosis 5. Nephrotic syndrome
Rare causes	5. Meig's syndrome 6. Constrictive pericarditis	6. Hypothyroidism

Encephalopathy

- Hepatic encephalopathy is brain dysfunction caused by liver insufficiency, portosystemic shunting, or both
- Ammonia, a toxin normally removed by the liver, plays a key role in its pathogenesis
- The presence of large amounts of ammonia in the brain damages supporting brain cells or astrocytes and leads to structural changes characteristic of hepatic encephalopathy (Alzheimer type II astrocytosis)
- occurs at a rate of approximately 2 to 3% per year
- gradual onset and rarely fatal

West Haven Grade For HE

Stage	Clinical features
0	No abnormalities
1	Alterations in behaviour Mild confusion Disordered sleep
2	Lethargy Moderate confusion Asterixis
3	Somnolent but arousable
4	Coma

Cardiopulmonary Complications

1. Cirrhotic cardiomyopathy (high output cardiac failure)
2. Hepatopulmonary syndrome
3. Portopulmonary hypertension

Hepato Pulmonary Syndrome

Symptoms and signs → Exertional dyspnea, clubbing, cyanosis

Pathophysiology :

- Vasodilation at the level of the pulmonary circulation leads to arterial hypoxemia.
- Normal pulmonary capillaries are 8 µm in diameter, and red blood cells (slightly less than 8 µm) pass through them one cell at a time, thereby facilitating oxygenation.
- In hepatopulmonary syndrome, the pulmonary capillaries are dilated up to 500 µm, so passage of red cells through the pulmonary capillaries may be many cells thick → large number of red cells are not oxygenated → equivalent of a right-to-left shunt

Hepato Pulmonary Syndrome

Diagnosis:

- arterial hypoxemia with a Pao2 of less than 80 mm Hg or an alveolar arterial oxygen gradient of greater than 15 mm Hg
- along with evidence of pulmonary vascular shunting on contrast echocardiography or a 99mTc-labeled macroaggregated albumin scan demonstrating abnormal shunting of radioactivity to the brain

Portopulmonary hypertension

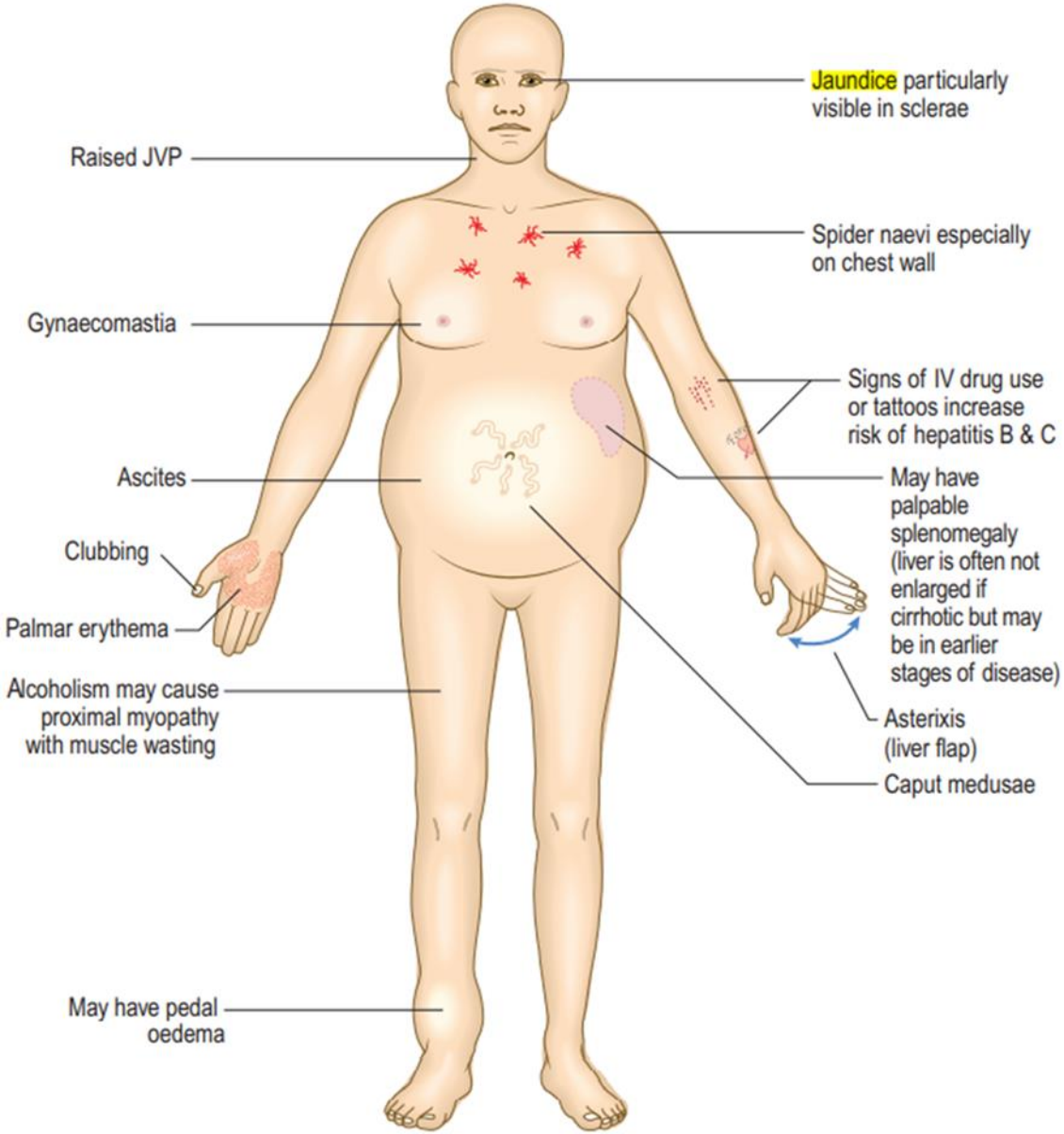
Symptoms and signs: Exertional dyspnea, syncope, chest pain, loud P2, RV heave

Pathophysiology:

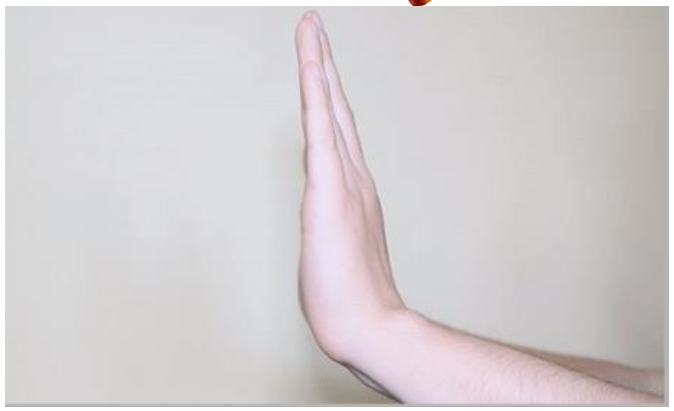
- Pulmonary bed is exposed to vasoconstrictive substances that may be produced in the splanchnic circulation and bypass metabolism by the liver
- Initially reversible pulmonary HTN
- Later irreversible pulmonary hypertension → endothelial proliferation, vasoconstriction, in situ thrombosis, and obliteration of vessels

Diagnosis: Mean pulmonary arterial pressure higher than 25 mm Hg on right-sided heart catheterization, provided pulmonary capillary wedge pressure is less than 15 mm Hg

General Examination



Signs Of Chronic Liver Disease

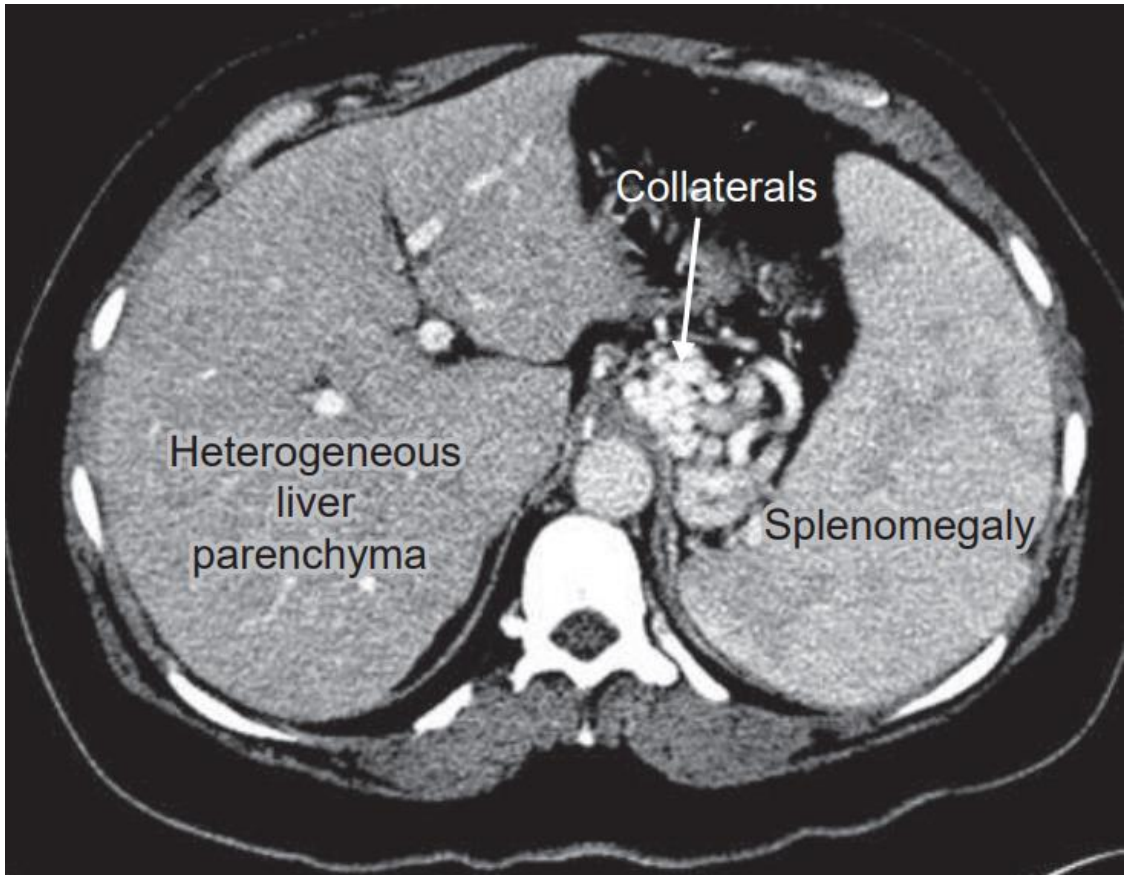


Laboratory Tests

- The most sensitive and specific laboratory finding suggestive of cirrhosis in the setting of chronic liver disease is a low platelet count (<1,50,000) which occurs as a result of portal hypertension and hypersplenism
- Decrease in albumin and increase in globulin → A/G reversal
- Elevation in bilirubin and liver enzymes
- Elevated PT INR

Imaging Studies

- USG, CT,MRI
- Nodular liver, small liver with or without hypertrophy of left or caudate lobe, splenomegaly, intrabdominal collaterals, ascites



- Transient elastography (Fibroscan)

  → assess stiffness of liver

Prognosis

TABLE 153-2 THE TWO MOST COMMONLY USED SCORING SYSTEMS IN CIRRHOSIS

1. CHILD-TURCOTTE-PUGH (CTP) SCORE (RANGE, 5-15)

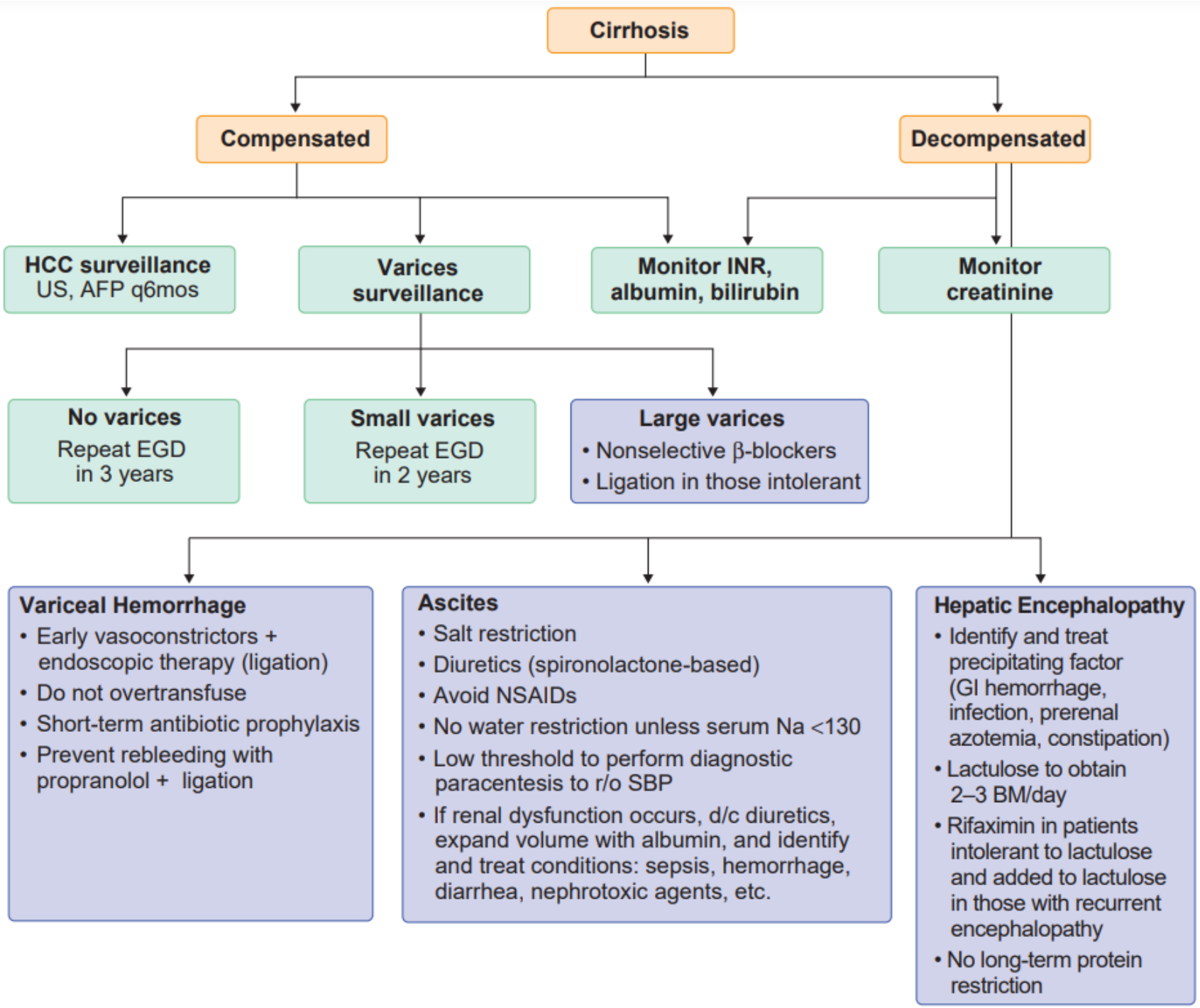
	POINTS ASCRIBED		
Parameters	**1**	**2**	**3**
Ascites	None	Grade 1-2 (or easy to treat)	Grade 3-4 (or refractory)
Hepatic encephalopathy	None	Grade 1-2 (or induced by a precipitant)	Grade 3-4 (or spontaneous)
Bilirubin (mg/dL)	<2	2-3	>3
Albumin (g/dL)	>3.5	2.8-3.5	<2.8
Prothrombin time (seconds > control) or INR	<4 <1.7	4-6 1.7-2.3	>6 >2.3

CTP classification: Child A: score of 5-6; Child B: score of 7-9; Child C: score of 10-15

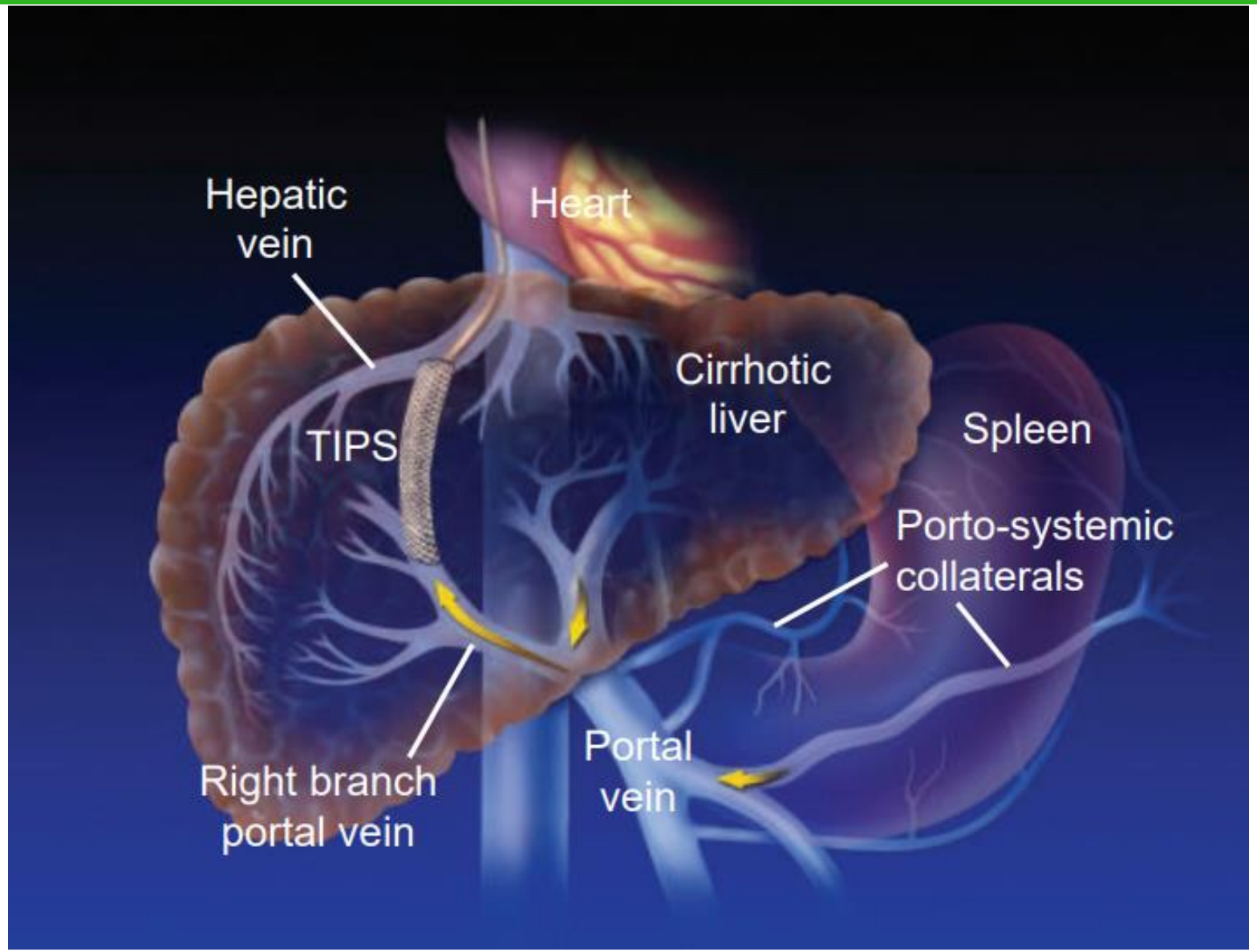
2. MODEL OF END-STAGE LIVER DISEASE (MELD) SCORE (RANGE, 6-40)

$[0.957 \times \text{LN (creatinine in mg/dL)} + 0.378 \times \text{LN (bilirubin in mg/dL)} + 1.12 \times \text{LN (INR)} + 0.643] \times 10$

INR = international normalized ratio; LN = natural logarithm.


Cirrhosis
Compensated
Decompensated
HCC surveillance
US, AFP q6mos
Varices surveillance
Monitor INR, albumin, bilirubin
Monitor creatinine
No varices
Repeat EGD in 3 years
Small varices
Repeat EGD in 2 years
Large varices
• Nonselective β-blockers
• Ligation in those intolerant
Variceal Hemorrhage
• Early vasoconstrictors + endoscopic therapy (ligation)
• Do not overtransfuse
• Short-term antibiotic prophylaxis
• Prevent rebleeding with propranolol + ligation
Ascites
• Salt restriction
• Diuretics (spironolactone-based)
• Avoid NSAIDs
• No water restriction unless serum Na <130
• Low threshold to perform diagnostic paracentesis to r/o SBP
• If renal dysfunction occurs, d/c diuretics, expand volume with albumin, and identify and treat conditions: sepsis, hemorrhage, diarrhea, nephrotoxic agents, etc.
Hepatic Encephalopathy
• Identify and treat precipitating factor (GI hemorrhage, infection, prerenal azotemia, constipation)
• Lactulose to obtain 2–3 BM/day
• Rifaximin in patients intolerant to lactulose and added to lactulose in those with recurrent encephalopathy
• No long-term protein restriction

TIPS


Hepatic vein
Heart
Cirrhotic liver
Spleen
TIPS
Porto-systemic collaterals
Portal vein
Right branch portal vein